

Final Report: M1/M2 Macrophage Contributions to Colon Cancer DTC Dormancy

Experimental Design with Workflow Diagrams and Bench Protocols

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Executive Summary

This report presents a comprehensive experimental plan to quantify the causal contribution of M1-like and M2-like macrophage phenotypes to dormancy entry, maintenance, and exit in colon cancer liver metastasis. The study is grounded in a multi-agent research analysis (FINAL_COMPILATION.md) that estimated M1 contributions at ~40% for entry and ~35% for exit, with M2 contributions at ~50% for maintenance and ~35% for exit. **These estimates are mechanistic inferences, not measurements. This experiment aims to generate the first direct quantitative data.**

Model: MC38 murine colon adenocarcinoma cells (GFP-Luciferase-Fucci2a reporters) injected intrasplenically into C57BL/6 mice to generate liver metastases.

Scale: 330 mice across 4 experimental arms, 7-month timeline, ~\$184,000 budget.

Key innovation: This is the first study to simultaneously quantify M1 and M2 contributions to each phase of dormancy (entry, maintenance, exit) using selective genetic and pharmacological macrophage manipulation.

Scientific Rationale and Interaction Model

Hypothesis

Both M1-like (classically activated, pro-inflammatory) and M2-like (alternatively activated, tissue-repair) macrophage phenotypes contribute to dormancy entry and exit in Disseminated Tumor Cells (DTCs), but through distinct mechanisms with different temporal profiles:

- **Dormancy entry:** M1 contribution ~40% (IFN-gamma/p38 MAPK/STAT1 axis), M2 contribution ~25% (TGF-beta2/GJIC)
- **Dormancy maintenance:** M1 ~15%, M2 ~50% (TGF-beta2, GJIC, niche architecture)
- **Dormancy exit:** M1 ~35% (NF-kappaB, inflammatory cytokines), M2 ~35% (angiogenic switch, MMPs, TGF-beta1/NRP2)

The diagram below illustrates the mechanistic model: M1-like macrophages DRIVE dormancy entry via IFN-gamma/p38/STAT1 signaling, M2-like macrophages MAINTAIN dormancy via TGF-beta2 and gap junctional intercellular communication (GJIC), and M2-like macrophages also TRIGGER exit through MMP-mediated ECM remodeling and angiogenic switching.

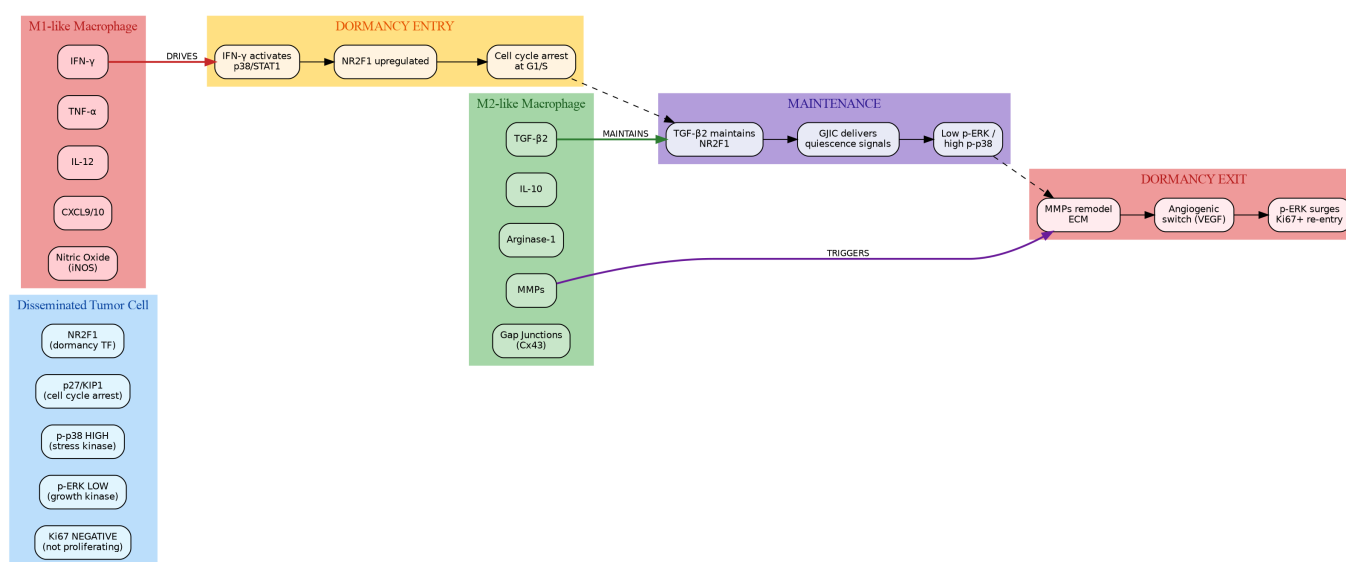


Figure 1: Mechanistic model of M1-driven dormancy entry (IFN-gamma/p38/STAT1), M2-driven maintenance (TGF-beta2/GJIC), and M2-triggered exit (MMPs/VEGF). DTC markers: NR2F1, p27, p-p38/p-ERK ratio.

Primary Objective

Quantify the causal contribution of M1-like and M2-like macrophages to each phase of dormancy (entry, maintenance, exit) in colon cancer DTCs using selective in vivo macrophage manipulation in a liver metastasis model.

Secondary Objectives

1. Identify macrophage-derived signaling molecules necessary and sufficient for dormancy entry vs. exit
2. Determine whether the same macrophage subpopulation maintains and prevents exit, or whether these are distinct subsets
3. Map the single-cell transcriptomic landscape of macrophage-DTC interactions during each dormancy phase
4. Test whether therapeutic M1 repolarization (PI3K-gamma inhibition or STING agonism) awakens dormant DTCs before clearing them

Model System and Dormancy Timeline

Cancer Cell Line: MC38

MC38 murine colon adenocarcinoma (C57BL/6 background) is the primary model, chosen for: - Compatibility with C57BL/6-based genetic tools (LysM-Cre, CD169-DTR) - Published protocols for splenic injection to liver metastasis - Immunocompetent syngeneic model with intact T cells, NK cells, and macrophages

Engineered reporters:

Reporter	Purpose
GFP/Luciferase	BLI tracking + flow cytometry identification
Fucci2a	Real-time cell cycle (red=G1 dormant, green=S/G2/M proliferating)
H2B-mCherry	Nuclear label for single-cell tracking

Intrasplenic Injection Timeline

The timeline below traces DTC fate from the moment of intrasplenic injection through seeding, dormancy entry, maintenance, and exit. On Day 0, 1×10^3 MC38-GFP-Luc-Fucci2a cells are injected into the spleen. Over Days 1-3, DTCs transit the portal vein and lodge in liver sinusoids. During Days 3-14, M1-like macrophage encounters drive ~65% of DTCs into dormancy (NR2F1+, Ki67-). M2-like macrophages then dominate the maintenance phase (Days 14-84). Spontaneous exit begins variably at Weeks 8-12.

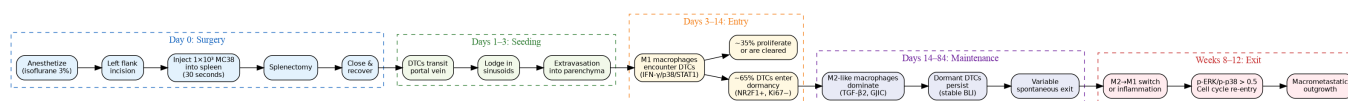


Figure 2: Intrasplenic injection and dormancy timeline. Surgery (Day 0) through seeding (Days 1-3), entry (Days 3-14), maintenance (Days 14-84), and exit (Weeks 8-12). Color-coded by phase.

Mouse Strains

Strain	Purpose
C57BL/6J	Wild-type control
LysM-Cre x Nos2fl/fl	M1 impairment (no iNOS in myeloid cells)
LysM-Cre x Arg1fl/fl	M2 impairment (no arginase-1 in myeloid cells)
CD169-DTR	Kupffer cell-specific depletion
Cx3cr1-CreERT2 x Rosa26-DTA	Recruited monocyte-derived macrophage depletion

Metastatic Niche: Liver

The liver is the primary metastatic site for colon cancer. Kupffer cells (liver-resident macrophages, ~80-90% of hepatic macrophages) are the first cells DTCs encounter upon arrival. The distinction between Kupffer cells and recruited monocyte-derived macrophages is critical for interpreting M1/M2 contributions.

Overall Project Workflow

The project follows a phased design: Phase 0 establishes reagents and dose-response data, Phase I confirms that macrophages are required (Arm A), Phase II quantifies M1 and M2 contributions separately (Arms B and C in parallel), Phase III tests the temporal switch hypothesis (Arm D), and Phase IV integrates all data. Decision gates at each phase determine whether to proceed or revise the model.

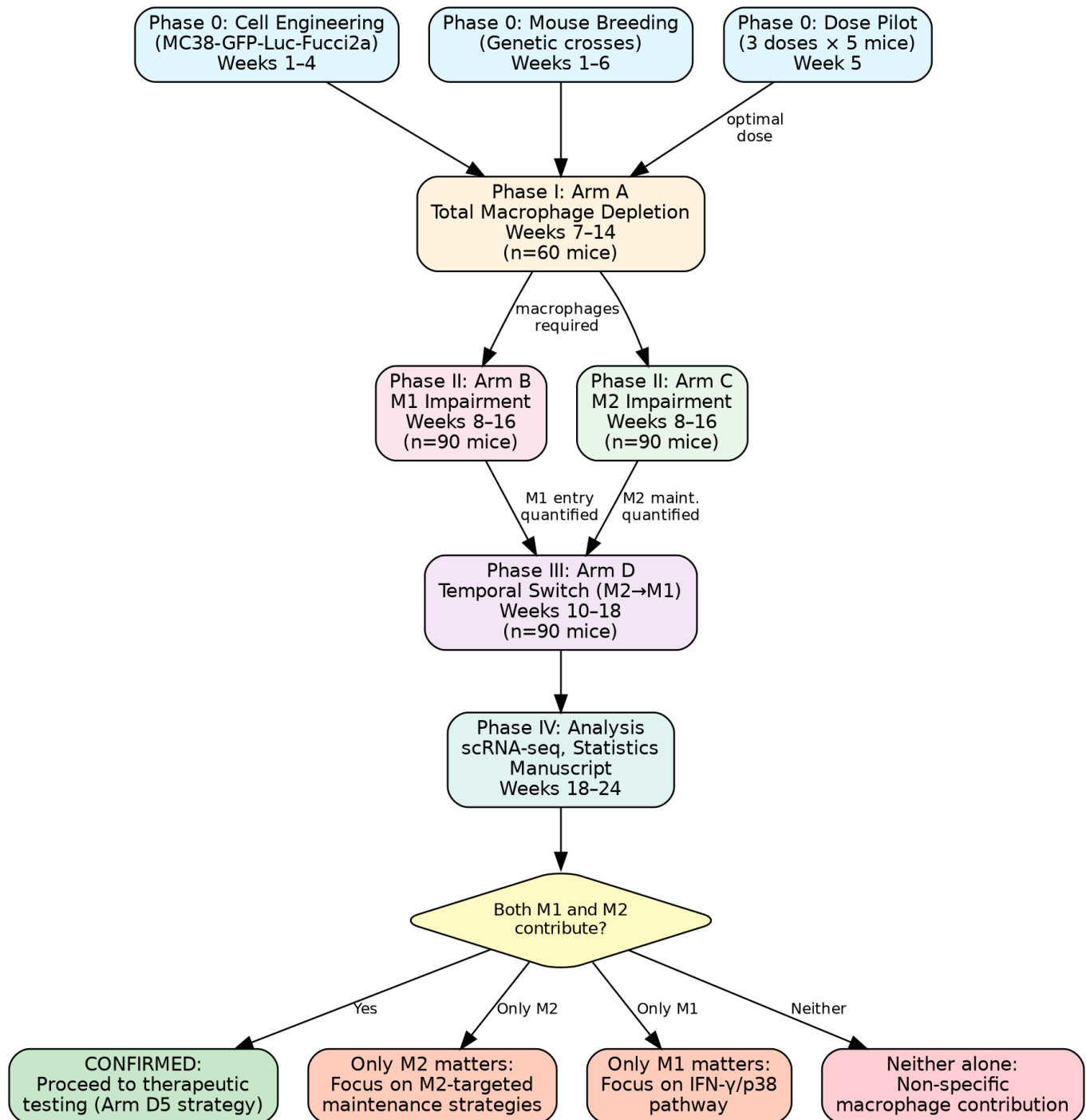


Figure 3: Overall project workflow from Phase 0 (cell engineering, mouse breeding, dose pilot) through Phase IV (analysis). Diamond decision node at bottom shows four possible outcome scenarios.

Experimental Design: Four Arms

Overview

The experiment consists of four arms, each testing a specific aspect of the M1/M2 contribution model. The diagram below summarizes all treatment groups, mouse numbers, and the research question each arm addresses.

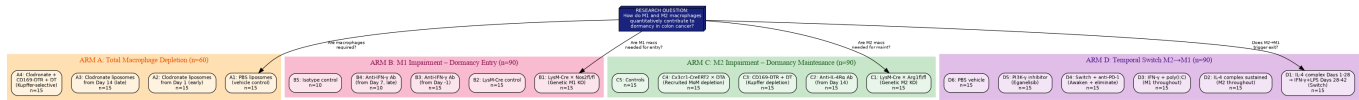


Figure 4: Four experimental arms: (A) Total macrophage depletion tests whether macrophages are required; (B) M1 impairment tests dormancy entry; (C) M2 impairment tests dormancy maintenance; (D) Temporal switch tests M2-to-M1 transition. n=330 mice total.

Arm A: Total Macrophage Depletion (n=60 mice)

Question: Are macrophages required for dormancy regulation?

Group	Treatment	Start	Expected
A1	PBS liposomes (control)	Day 1	Dormant DTCs persist
A2	Clodronate liposomes (early)	Day 1	Reduced entry or maintenance
A3	Clodronate liposomes (late)	Day 14	Impaired maintenance
A4	Clodronate + CD169-DTR + DT	Day 14	Kupffer-specific effect

Protocol: 200 uL IV or IP twice weekly. Verify >80% depletion by flow cytometry.

Arm B: M1 Impairment (n=90 mice)

Question: Are M1 macrophages required for dormancy entry?

Group	Genotype/Treatment	Expected
B1	LysM-Cre x Nos2fl/fl	Entry rate: 65% to ~40%
B2	LysM-Cre control	No change
B3	Anti-IFN-gamma Ab (early)	Entry rate: ~40%
B4	Anti-IFN-gamma Ab (late, Day 7)	Maintenance unaffected
B5	Isotype controls	Baseline

Primary endpoint: Dormancy entry rate (Ki67- NR2F1+ GFP+ cells / total GFP+) at Day 7.

Arm C: M2 Impairment (n=90 mice)

Question: Are M2 macrophages required for dormancy maintenance?

Group	Genotype/Treatment	Expected
C1	LysM-Cre x Arg1fl/fl	Maintenance: 8wk to ~4wk
C2	Anti-IL-4Ralpha Ab (from Day 14)	Maintenance: ~5wk
C3	CD169-DTR + DT (from Day 14)	Kupffer contribution
C4	Cx3cr1-CreERT2 x DTA	Recruited MoM contribution
C5	Controls	Baseline 8wk

Primary endpoint: Dormancy maintenance duration (Kaplan-Meier, log-rank test).

Arm D: Temporal Switch (n=90 mice)**Question:** Does M2-to-M1 switching trigger dormancy exit?

Group	Treatment	Expected
D1	IL-4 complex -> IFN-gamma+LPS (switch)	Exit at ~3wk post-switch
D2	IL-4 complex sustained (M2)	Prolonged dormancy >10wk
D3	IFN-gamma + poly(I:C) sustained (M1)	Early inflammatory exit ~4wk
D4	Switch + anti-PD-1	Exit + clearance
D5	PI3K-gamma inhibitor (eganelisib)	Gradual repolarization ~5wk
D6	PBS vehicle	Natural kinetics

Critical comparison: D4 vs D1 - does anti-PD-1 convert “awakening” into “clearance”?

Readout Hierarchy and Tissue Processing

Three-Tier Readout System

Readouts are organized in three tiers: primary (mandatory for all mice), secondary (subset), and tertiary (dedicated cohort). All tiers feed into a unified computational analysis pipeline producing four quantitative metrics: Dormancy Entry Rate (DER), Dormancy Maintenance Duration (DMD), Dormancy Exit Rate (DXR), and DTC Clearance Rate (DCR).

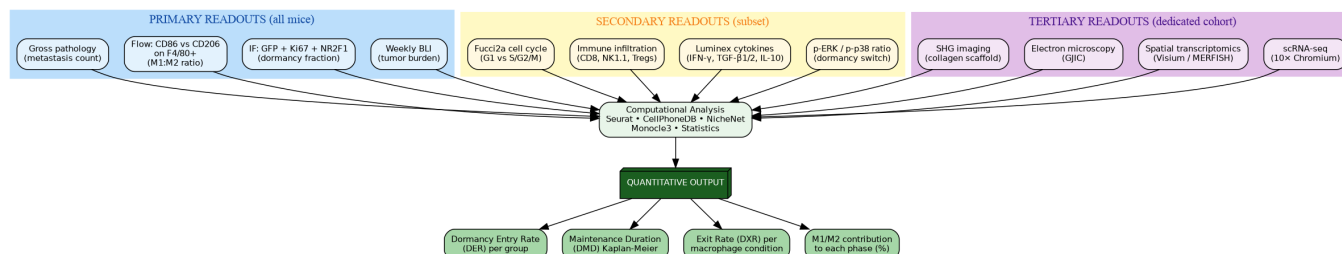
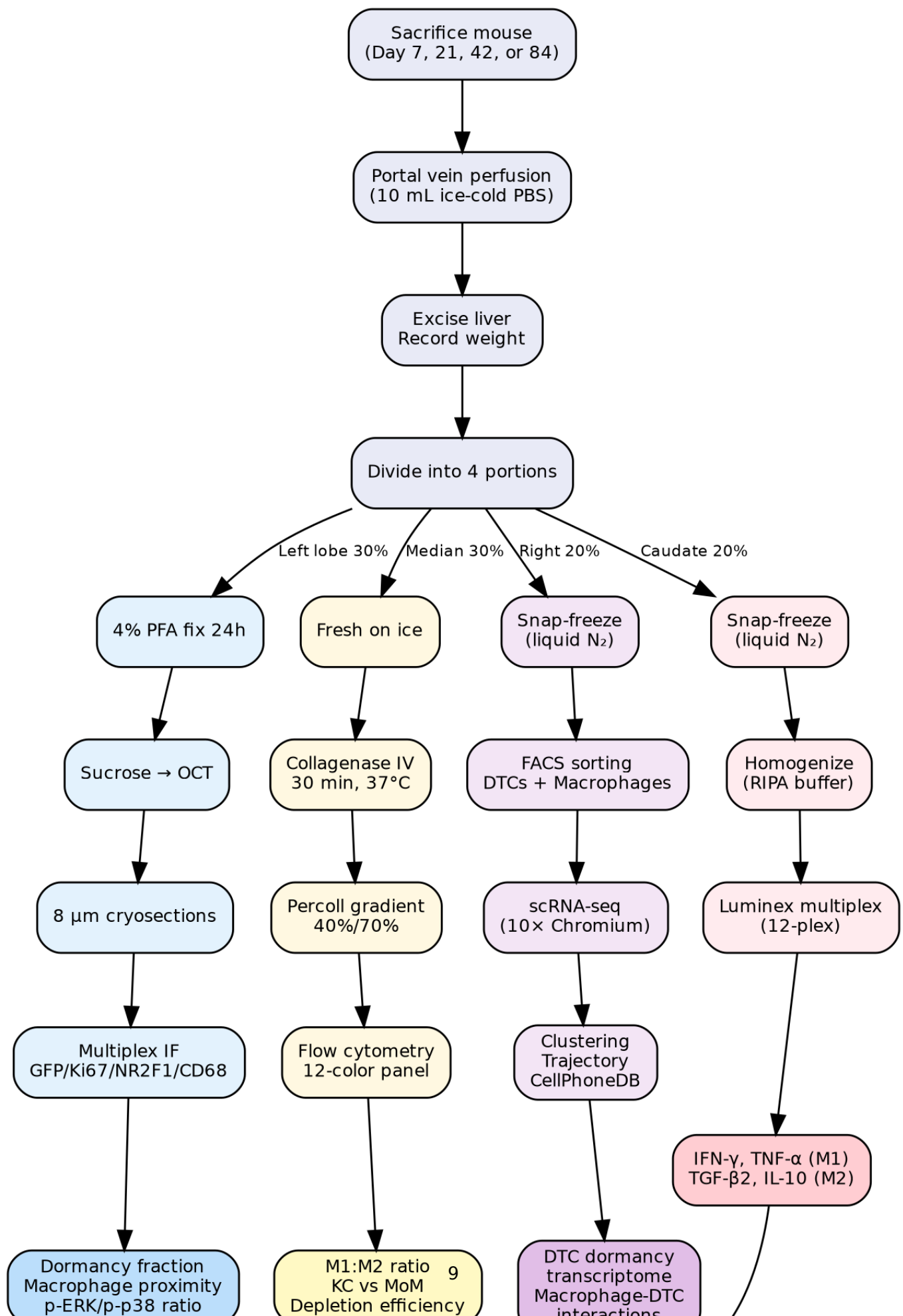


Figure 5: Three-tier readout hierarchy. Primary: BLI, IF, flow cytometry, gross pathology. Secondary: p-ERK/p-p38, cytokines, immune infiltration, Fucci2a. Tertiary: scRNA-seq, spatial transcriptomics, EM, SHG imaging. All feed into computational analysis.

Tissue Processing Pipeline

At each sacrifice timepoint (Day 7, 21, 42, 84), the liver is divided into four portions for parallel processing. The pipeline below shows how each portion is processed for immunofluorescence, flow cytometry, scRNA-seq, or cytokine analysis, with all data streams integrating into a per-mouse quantitative report.



Expected Outcomes and Decision Framework

Predicted Results

Arm	Key Group	Predicted Entry Rate	Predicted Maintenance
A	A1 (control)	~65%	~8 weeks
A	A2 (all depleted, early)	~30%	~3 weeks
B	B1 (M1-impaired)	~40%	~7 weeks
C	C1 (M2-impaired)	~60%	~4 weeks
D	D1 (M2->M1 switch)	~65%	Exit at ~3wk post-switch
D	D2 (sustained M2)	~65%	>10 weeks
D	D4 (switch + anti-PD-1)	~65%	Exit + clearance

Decision Tree

The decision tree below shows how experimental results will validate or refute the Temporal Switch Model. The central question is whether both M1 and M2 contribute to dormancy regulation. If confirmed (Scenario 1), the Arm D5/D4 anti-PD-1 combination strategy is tested as a clinical translation pathway. If only one phenotype matters, the model is revised accordingly.

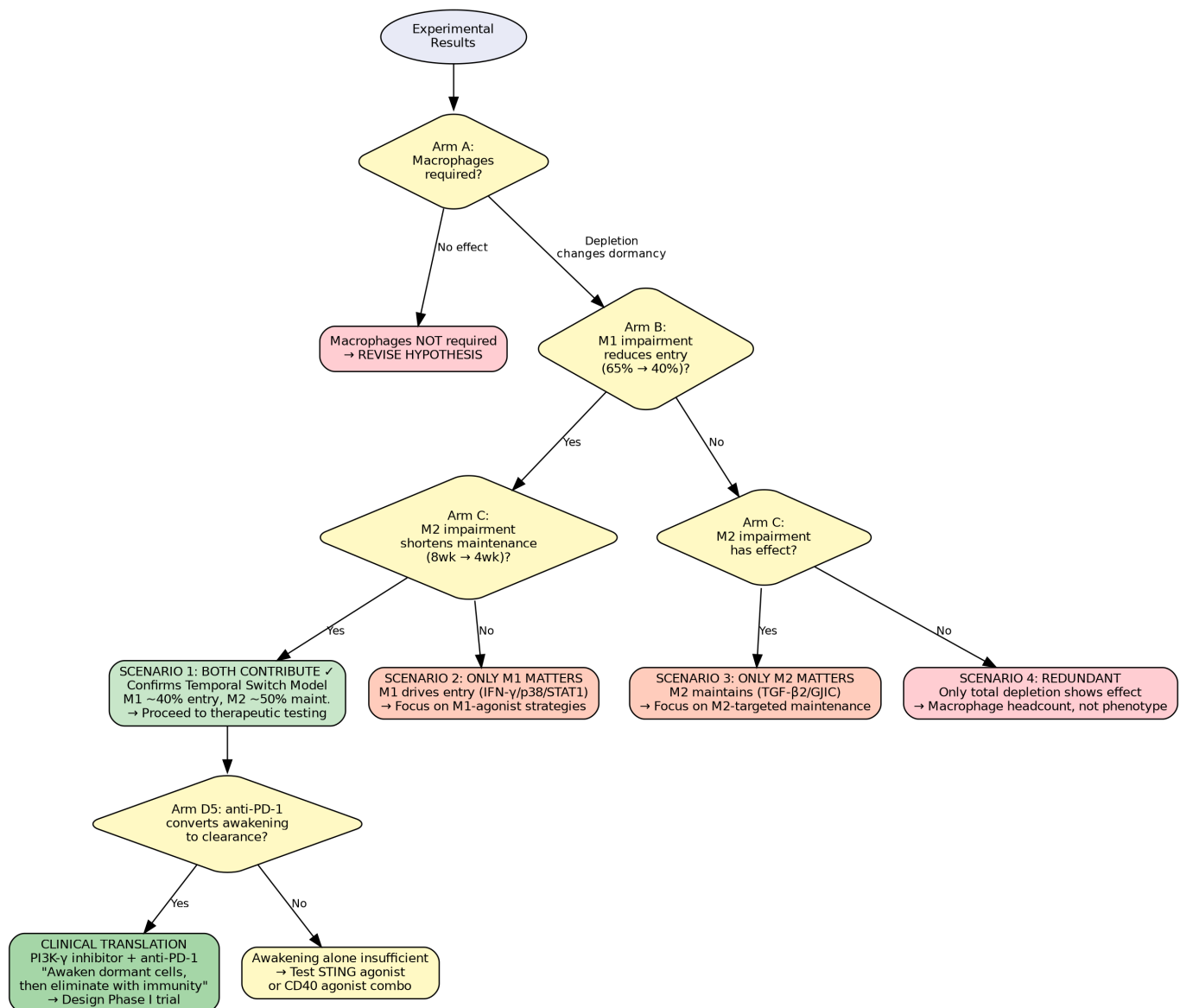


Figure 7: Expected outcomes decision tree. Four scenarios based on Arm A-D results: (1) Both M1 and M2 contribute (confirms model, proceed to therapy); (2) Only M1 (focus on IFN-gamma/p38); (3) Only M2 (focus on TGF-beta2/GIIC); (4) Neither alone (non-specific contribution). Green = confirmed model; orange/red = model revision required.

Decision Criteria

Scenario 1 (Both contribute): B1 shows reduced entry AND C1 shows reduced maintenance. D4 shows clearance. -> Proceed to therapeutic testing. Design Phase I trial (PI3K-gamma inhibitor + anti-PD-1).

Scenario 2 (Only M2): B groups show no effect; only C groups change dormancy. -> Revise model. Focus on M2-targeted maintenance strategies.

Scenario 3 (Only M1): C groups show no effect; only B groups affect dormancy. -> Revise model. Focus on IFN-gamma/p38 pathway.

Scenario 4 (Neither alone): Only total depletion (Arm A) shows phenotype. -> Macrophage contribution is non-specific (headcount, not phenotype).

Statistical Framework

Power Analysis

Comparison	Effect Size	Alpha	Power	n per group
Dormancy entry rate (Arm B)	65% vs 40% (25% diff)	0.05	0.80	15
Maintenance duration (Arm C)	HR=2.0 (8wk vs 4wk)	0.05	0.80	15
Temporal switch (Arm D)	3wk vs 8+ wk	0.05	0.80	15

Statistical Tests

Comparison	Test	Correction
Dormancy fraction	Fisher's exact	Bonferroni (6 arms)
Time to metastasis	Kaplan-Meier + log-rank	Bonferroni
p-ERK/p-p38 ratio	Mann-Whitney U	Holm-Bonferroni
scRNA-seq DEGs	Wilcoxon rank-sum	Benjamini-Hochberg FDR
Cytokine levels	One-way ANOVA + Tukey	Tukey HSD
Longitudinal BLI	Mixed-effects model	FDR

Total: 330 mice across 4 arms, 7-month timeline, ~\$184,000.

Key Reagent and Vendor Summary

Reagent	Vendor	Catalog	Dose/Schedule
Clodronate liposomes	Liposoma BV	CP-005-005	200 uL IV, 2x/week
PBS liposomes	Liposoma BV	PB-005-005	200 uL IV, 2x/week
CpG-ODN 1826	InvivoGen	tlrl-1826	50 ug IP, 3x/week
Recombinant mouse IL-4	BioLegend	574302	5 ug + 25 ug anti-IL-4 (complex)
Anti-IFN-gamma (XMG1.2)	Bio X Cell	BP0055	200 ug IP, 2x/week
Diphtheria toxin	Sigma	D0564	4 ng/g IP, every 72h
D-Luciferin	GoldBio	LUCK-1G	150 mg/kg IP
Pexidartinib (PLX3397)	Selleckchem	S7818	290 mg/kg chow
Eganelisib (IPI-549)	Selleckchem	S8330	30 mg/kg oral daily
Anti-PD-1 (clone RMP1-14)	Bio X Cell	BP0146	200 ug IP, 2x/week

Acronym Glossary

Acronym	Full Term
ARG1	Arginase 1
BLI	Bioluminescence Imaging
BMDM	Bone Marrow-Derived Macrophage
CRC	Colorectal Cancer
CSF1R	Colony Stimulating Factor 1 Receptor
CX3CR1	C-X3-C Motif Chemokine Receptor 1
DER	Dormancy Entry Rate
DMD	Dormancy Maintenance Duration
DT	Diphtheria Toxin
DTC	Disseminated Tumor Cell
DXR	Dormancy Exit Rate
ERK	Extracellular Signal-Regulated Kinase
Fucci2a	Fluorescent Ubiquitination-based Cell Cycle Indicator 2a
GJIC	Gap Junctional Intercellular Communication
IF	Immunofluorescence
IFN-gamma	Interferon gamma
IL	Interleukin
iNOS	Inducible Nitric Oxide Synthase (NOS2)
KC	Kupffer Cell
MMP	Matrix Metalloproteinase
MoM	Monocyte-derived Macrophage
NR2F1	Nuclear Receptor Subfamily 2 Group F Member 1
NRP2	Neuropilin 2
PD-1	Programmed Cell Death Protein 1
PI3K-gamma	Phosphoinositide 3-Kinase Gamma
scRNA-seq	Single-Cell RNA Sequencing
TAM	Tumor-Associated Macrophage
TGF-beta	Transforming Growth Factor Beta
TLR	Toll-Like Receptor